Week 1 - Graded Assignment 1

The due date for submitting this assignment has passed.
Due on 2026-02-18, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-02-18, 08:57 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ):

1 point
Match the matrices in column A with their properties in column B and answer the following question.

Which of the following are true?

- [x] a) → → iii)
- [ ] a) → → ii)
- [x] b) → → i)
- [ ] b) → → ii)
- [x] c) → → iv)
- [ ] c) → → iii)
- [ ] d) → → iv)
- [x] d) → → ii)

Yes, the answer is correct.
Score: 1
Accepted Answers:
a) → → iii)
b) → → i)
c) → → iv)
d) → → ii)

1 point
Match the systems of linear equations in Column A with their number of solutions in column B and their geometric representation in Column C and answer the following question.

Which of the following are true?

- [x] i) → → b) → → 3)
- [ ] i) → → a) → → 3)
- [ ] ii) → → c) → → 2)

- [x] ii) → → c) → → 1)
- [x] iii) → → a) → → 2)
- [ ] iii) → → a) → → 1)

Yes, the answer is correct.

Score: 1

Accepted Answers:

i) → → b) → → 3)
ii) → → c) → → 1)
iii) → → a) → → 2)

1 point

Let $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$B = $\begin{bmatrix} & \\ & \\ 1 & 0 -1 0 1 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 0 \\ -1 & 2 \end{bmatrix}$C = [0−102]. Which of the following options are true for a matrix AA, such that $AB = C$AB = C?

- [ ] Such a matrix does not exist.
- [ ] There is a unique matrix AA satisfying this property.
- [x] There are infinitely many such matrices.
- [x] AA should be a $2 \times 3$2 × 3 matrix.
- [ ] AA should be a $3 \times 2$3 × 2 matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

There are infinitely many such matrices.
AA should be a $2 \times 3$2 × 3 matrix.

1 point

Let AA be a $2 \times 2$2 × 2 real matrix and let $trace(A)$trace(A) denote the sum of the elements in the diagonal of AA. Which of the following are true?

- [ ] $det(A - cI)$det(A − cI) is a polynomial in cc of degree 1.
- [ ] $det(A - cI)$det(A − cI) is a polynomial in cc of degree 2.
- [x] $det(A - cI) = c^2 - trace(A)c + det(A)$det(A − cI) = c2 − trace(A)c + det(A)
- [ ] $det(A - cI) = c^2 + trace(A)c - det(A)$det(A − cI) = c2 + trace(A)c − det(A)
- [ ] $det(A - cI) = trace(A)c - det(A)$det(A − cI) = trace(A)c − det(A)
- [ ] $det(A - cI) = -trace(A)c + det(A)$det(A − cI) = −trace(A)c + det(A)

Partially Correct.

Score: 0.5

Accepted Answers:

$det(A - cI)$det(A − cI) is a polynomial in cc of degree 2.
$det(A - cI) = c^2 - trace(A)c + det(A)$det(A − cI) = c2 − trace(A)c + det(A)

1 point

Suppose there are two types of oranges and two types of bananas available in the market. Suppose 1 kg of each type of orange costs ₹50 and 1 kg of each type of banana costs ₹40. Gargi bought xx kg of the first type of each fruit, orange and banana, and yy kg of the second type of each fruit, orange and banana. She paid ₹250 for oranges and ₹200 for bananas. Which of the following options are correct with respect to the given information?

- [x] The matrix representation to find xx and yy can be $\begin{bmatrix}50 & 50\\ 40 & 40\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}250\\ 200\end{bmatrix}$[50405040] [xy] = [250200]
- [ ] The matrix representation to find xx and yy can be $\begin{bmatrix}50 & 40\\ 50 & 40\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}250\\ 200\end{bmatrix}$[50504040] [xy] = [250200]
- [ ] The matrix representation to find xx and yy can be $\begin{bmatrix}40 & 40\\ 50 & 50\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}200\\ 250\end{bmatrix}$[40504050] [xy] = [200250]
- [x] xx can be 2 and yy can be 3.
- [x] There are infinitely many real values possible for xx and yy.
- [ ] There are only finitely many real values possible for xx and yy.
- [ ] There are only finitely many natural numbers possible for xx and yy.

Partially Correct.

Score: 0.6

Accepted Answers:

The matrix representation to find xx and yy can be $\begin{bmatrix}50 & 50\\ 40 & 40\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}250\\ 200\end{bmatrix}$[50405040] [xy] = [250200]

The matrix representation to find xx and yy can be $\begin{bmatrix}40 & 40\\ 50 & 50\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}200\\ 250\end{bmatrix}$[40504050] [xy] = [200250]

xx can be 2 and yy can be 3.

There are infinitely many real values possible for xx and yy.

There are only finitely many natural numbers possible for xx and yy.

Numerical Answer Type (NAT):

Suppose $det(4A) = n \times det(A)$det(4A) = n × det(A) for any $5 \times 5$5 × 5 real matrix AA. What is the value of nn?

1024

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 1024

1 point

A, BA, B are two square matrices of the same order. If det(A) = 2det(A) = 2 and det(B) = 3det(B) = 3, what is det($A^2 B^3$)det(A2B3)?

108

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 108

1 point

Let AA be a square matrix of order 3 and BB be a matrix that is obtained by adding 88 times the first row of AA to the third row of AA and adding 99 times the second row of AA to the first row of AA. If $det(A)$ = 5det(A) = 5, then find out the value of $det(8A^2 B^{-1})$det(8A2B−1).

2560

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2560

1 point

Comprehension Type Question:

The scores of a student in mathematics, physics and chemistry in her class-12 board exams are m, pm, p and cc respectively, where each score is out of 100100. She has applied for three engineering streams in a college. Each stream assigns different weights to these three subjects to calculate her final score, which is again out of 100100.

For example, the weight given to mathematics, physics and chemistry could be 0.2, 0.70.2, 0.7 and 0.1 0.1 respectively by a stream. These weights are then multiplied with the corresponding scores to get the final score. Concretely, if the student has scored 85, 78, 4085, 78, 40 in the three subjects, her final score for this stream is:

0.2 × 85 + 0.7 × 78 + 0.1 × 40 = 75.60.2 × 85 + 0.7 × 78 + 0.1 × 40 = 75.6

This is called a weighted average. Note that the weights always sum to 11. Now, the weights assigned by the three streams for mathematics, physics and chemistry, in this order, are given below:

Stream-1: 0.2, 0.7, 0.10.2, 0.7, 0.1

Stream-2: 0.5, 0.3, 0.20.5, 0.3, 0.2

Stream-3: 0.1, 0.4, 0.50.1, 0.4, 0.5

The final score of the student in stream-1 is 8181. It is 8383 in stream-2 and 7676 in stream-3. We wish to find the student's marks in the three subjects.

1 point

This is framed as a system of linear equations. Select all true options concerning the coefficient matrix if the vector of unknowns is given as $\begin{bmatrix} m \\ p \\ c \end{bmatrix}$ $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$mpc. Assume that the first equation corresponds to stream-1, second to stream-2 and last to stream-3.

- [x] The first row is 0.2, 0.7, 0.10.2, 0.7, 0.1
- [x] The last row is 0.1, 0.4, 0.50.1, 0.4, 0.5
- [ ] The first column is 0.2, 0.7, 0.10.2, 0.7, 0.1

The middle column is 0.5, 0.3, 0.20.5, 0.3, 0.2

Yes, the answer is correct.

Score: 1

Accepted Answers:

The first row is 0.2, 0.7, 0.10.2, 0.7, 0.1

The last row is 0.1, 0.4, 0.50.1, 0.4, 0.5

Find $\frac{m+p+c}{3}$3m + p + c.

80

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 80

1 point

Find det(A)det(A), where AA is the coefficient matrix.

-0.13

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -0.13

1 point

Let A be a (4×4)(4 × 4) matrix such that

$(A + 3I)(A - 3I) = 0$(A + 3I)(A − 3I) = 0,

and suppose that $det(A) > 0$det(A) > 0. Find $det(A)$det(A).

81

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 81

1 point